CS 3100C: Honors Discrete Math

Lecture 19: Counting II

1 Who Gets the Snacks?

Before counting a distribution, decide whether the objects are *distinguishable* and whether a recipient may get none. Alice, Bob, and Charlie are different people, so swapping their shares usually changes the outcome.

Six different snacks. Suppose the six snacks are labeled $s_1, \ldots, s_6$ and every snack goes to exactly one of Alice, Bob, and Charlie. Each snack has three independent choices of recipient, so there are

$$\boxed{3^6 = 729}$$

distributions. Anyone may receive zero snacks. The 3^6 assignments count distributions, not orders in which the snacks are handed out.

Six identical cookies, with one for everyone. Now only the numbers of cookies received matter. One possible allocation is shown below.



Writing (x_A, x_B, x_C) for the three shares, we need $x_A + x_B + x_C = 6$ with each share at least 1. The possible shares are

$$(1,1,4),\ (1,2,3),\ (1,3,2),\ (1,4,1),$$
$$(2,1,3),\ (2,2,2),\ (2,3,1),$$
$$(3,1,2),\ (3,2,1),\ (4,1,1).$$

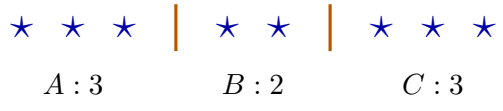
We will see how to count these solutions systematically.

2 Stars and Bars: Identical Objects

How many ways can we distribute eight identical cookies among Alice, Bob, and Charlie, allowing someone to receive none? We seek the nonnegative integer solutions of

$$x_A + x_B + x_C = 8.$$

Write eight stars in a row, then insert two bars to divide the row into Alice's, Bob's, and Charlie's shares. For example:



The bars may be adjacent or at either end: $||********$ represents $(0, 0, 8)$. Every allocation produces exactly one string of eight stars and two bars, and reading the numbers of stars between the bars recovers the allocation. This is a bijection. Choose the two bar positions among ten positions, giving

$$\boxed{\binom{8+2}{2} = \binom{10}{2} = 45}.$$

Principle 1 (Stars and bars)**.** *The number of nonnegative integer solutions of* $x_1 + \cdots + x_k = n$ *is*

$$\boxed{\binom{n+k-1}{k-1}}.$$

There are n *stars and* $k-1$ *bars. If all* $x_i \geq 1$*, first give one object to each person; then distribute the remaining* $n-k$ *objects. Thus, for* $n \geq k$*, the positive-solution count is* $\boxed{\binom{n-1}{k-1}}$.

For the six-cookie example, giving one cookie to each person leaves three cookies to distribute freely, so the count is $\binom{3+3-1}{3-1} = \binom{5}{2} = 10$.

3 Constrained Counting Problems

Problem 1. How many nonnegative integer triples satisfy $x + y + z \leq 10$?

Answer. The inequality becomes an equation when we introduce the unused amount $w = 10 - (x + y + z)$. Since $w \geq 0$, every allowed triple gives exactly one nonnegative solution of $x + y + z + w = 10$, and every such solution gives an allowed triple. Stars and bars with ten stars and three bars gives $\binom{10+4-1}{4-1} = \boxed{286}$.

Problem 2. How many ways can twelve identical cookies be distributed among four people if everyone receives at least one?

Answer. Give one cookie to each person first. This uses four cookies and removes the minimum requirement. The remaining eight can go to the four people in any nonnegative shares, so the answer is $\binom{8+4-1}{4-1} = \boxed{165}$.

Problem 3. How many ways can twelve identical cookies be distributed among Alice, Bob, and Charlie if Alice gets at least 2, Bob at least 3, and Charlie at least 1?

Answer. Set aside those required shares, which use $2 + 3 + 1 = 6$ cookies. The leftover shares $y_A = x_A - 2$, $y_B = x_B - 3$, and $y_C = x_C - 1$ are nonnegative and satisfy $y_A + y_B + y_C = 6$. Thus stars and bars gives $\binom{6+3-1}{3-1} = \boxed{28}$.

Problem 4. Twelve identical cookies must be given to four people in two-cookie units. How many distributions are possible if someone may receive none?

Answer. Think of the cookies as six identical pairs. If p_i is the number of pairs given to person i, then $p_1 + \cdots + p_4 = 6$ with each $p_i \geq 0$. The number of solutions is $\binom{6+4-1}{4-1} = \boxed{84}$.

Problem 5. How many ways can twelve identical cookies be distributed among four students if exactly two students receive cookies?

Answer. Choose the two recipients in $\binom{4}{2}$ ways. Their shares must both be positive and sum to 12. The first chosen student may receive any of $1, 2, \ldots, 11$ cookies; the other receives the rest. Hence the answer is $\binom{4}{2} \cdot 11 = \boxed{66}$.

Problem 6. How many ways can eight identical cookies be distributed among Alice, Bob, and Charlie if Alice receives at most three? Give a count by cases and a count by complement.

Answer. *By cases:* If Alice gets a cookies, Bob and Charlie split the other $8 - a$ in $9 - a$ ways. For $a = 0, 1, 2, 3$, the counts are $9, 8, 7, 6$, so the total is $\boxed{9 + 8 + 7 + 6 = 30}$.

By complement: There are $\binom{10}{2} = 45$ unrestricted distributions. If Alice gets at least four, remove four from her share; the remaining four cookies have $\binom{6}{2} = 15$ distributions among three people. Subtracting gives $\boxed{45 - 15 = 30}$.

Problem 7. How many ways can eight identical cookies be distributed among three students if nobody receives more than four?

Answer. Start with all $\binom{10}{2} = 45$ distributions. For a specified student to violate the bound, give that student five cookies first; the remaining three have $\binom{5}{2} = 10$ distributions. Two students cannot both receive at least five from only eight cookies, so these three sets of violations do not overlap. The answer is $45 - 3\binom{5}{2} = \boxed{15}$.

Problem 8. How many length-10 binary strings have exactly four 1s, with no two 1s adjacent?

Answer. Place one required 0 between each consecutive pair of 1s. This uses three of the six zeros. The remaining three zeros can be distributed among five gaps: before the first 1, in the three internal gaps, and after the last 1. Stars and bars gives $\binom{3+5-1}{5-1} = \boxed{35}$. Equivalently, if the positions of the 1s are $p_1 < \cdots < p_4$, then $p_1, p_2 - 1, p_3 - 2, p_4 - 3$ form a four-element subset of $\{1, \ldots, 7\}$.

Problem 9. Twelve different cookies come in six *fixed* pairs, such as $(c_1, c_2), \ldots, (c_{11}, c_{12})$. Each pair must stay together and go to one of four people. How many distributions are possible?

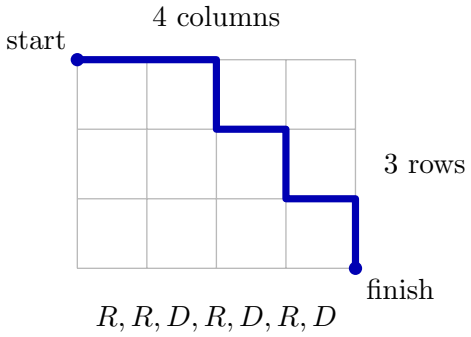
Answer. The six pairs are distinguishable units. Each unit has four independent choices of recipient, giving $\boxed{4^6 = 4{,}096}$. We do not count ways to form pairs: the pairings are already specified. For example, sending (c_1, c_2) to Alice is one assignment, regardless of which cookie is handed over first.

Problem 10. What if the twelve different cookies must first be paired, and the pairing is recorded as part of the outcome?

Answer. There are $12!/(2^6 6!) = 10{,}395$ ways to partition the cookies into six unordered pairs. The factor 2^6 removes the order within each pair; $6!$ removes the order of the six pairs. Each resulting pair then has four choices of recipient, so the count is $\boxed{10{,}395 \cdot 4^6 = 42{,}577{,}920}$. This counts the pairings as well as the assignments. If we record only each student's final pile, multiplying by the number of pairings would overcount: four cookies in one pile can be paired three ways without changing that pile.

4 Grid Paths: The Same Binomial Coefficients

Consider a grid of four columns and three rows of unit squares. Start at the upper-left corner and move only right (R) or down (D) to the lower-right corner. Each route needs four right moves to cross the columns and three down moves to cross the rows. The picture shows the route $RRDRDRD$.



The move sequence determines every turn of a route. Conversely, any sequence with four Rs and three Ds reaches the lower-right corner and cannot leave the grid. So counting routes is exactly counting these seven-letter sequences.

Method 1: permute repeated moves. Temporarily label the right moves $R_1, \ldots, R_4$ and the down moves $D_1, \ldots, D_3$. The seven labeled moves have $7!$ orders. Once the labels are erased, each visible route has been counted $4!3!$ times: the four R labels and three D labels may be permuted without changing the route. Therefore

$$\boxed{\frac{7!}{4!\,3!} = 35}.$$

Method 2: choose move positions. Number the seven move slots $1, \dots, 7$. Choose the three slots for D; every other slot must be R. For the pictured route the Ds occupy slots $\{3, 5, 7\}$. This choice uniquely determines the route, and every route gives one such choice. Hence

$$\boxed{\binom{7}{3} = 35}.$$

We could instead choose the four R slots, giving $\binom{7}{4} = 35$. Thus the factorial formula and the binomial coefficient describe the same count: $\binom{7}{3} = 7!/(3!4!)$.

4.1 Counting paths by dynamic programming

Let $F(r, c)$ count routes to the vertex r steps down and c steps right. A vertex on the top or left edge has just one route, so $F(0, c) = F(r, 0) = 1$. Every interior route enters from above or from the left, in disjoint cases:

$$\boxed{F(r, c) = F(r-1, c) + F(r, c-1) \qquad (r, c > 0).}$$

Fill the table in order from top to bottom and left to right:

	$c = 0$	1	2	3	4
$r = 0$	1	1	1	1	1
1	1	2	3	4	5
2	1	3	6	10	15
3	1	4	10	20	$\boxed{35}$

At the destination, $F(3, 4) = 20 + 15 = 35$. A route to (r, c) chooses its r down moves among $r + c$ positions, so $F(r, c) = \binom{r+c}{r}$. The table's rule is Pascal's identity.

5 Pascal's Identity and Pascal's Triangle

Theorem 1 (Pascal's identity)**.** *For integers $n \geq 1$ and $1 \leq k \leq n-1$,*

$$\boxed{\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}}.$$

Combinatorial proof. Count the k-element subsets of an n-element set after choosing one distinguished element e. Subsets containing e use $k-1$ of the remaining $n-1$ elements, giving $\binom{n-1}{k-1}$. Subsets without e use k of them, giving $\binom{n-1}{k}$. These cases are disjoint and exhaustive.

The identity also gives the grid recurrence, since the two terms count routes entering from above and from the left:

$$\binom{r+c}{r} = \binom{r+c-1}{r-1} + \binom{r+c-1}{r}.$$

Start with 1 at the top and edges, then add adjacent entries above to build *Pascal's triangle*:

						1								
					1		1							
				1		2		1						
			1		3		3		1					
		1		4		6		4		1				
	1		5		10		10		5		1			
1		6		15		20		15		6		1		
1		7		21		35		35		21		7		1

Row n contains $\binom{n}{0}, \binom{n}{1}, \ldots, \binom{n}{n}$. Row 7 includes our grid-path count $\binom{7}{3} = \binom{7}{4} = 35$.